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A MINI PROJECT
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DECEPTIVE UI & DARK PATTERN DETECTOR IN WEB/E-COMMERCE INTERFACES USING NLP AND OCR

1. ABSTRACT

E-commerce platforms and websites frequently employ manipulative interface strategies, known as “dark patterns,” to influence user decisions through artificial urgency, fabricated scarcity, guilt-tripping prompts, and misleading social-proof cues. This project presents a lightweight, real-time auditing tool that detects and classifies such deceptive text patterns from both raw text and UI screenshots. The system accepts dual-mode input: directly pasted text or uploaded interface screenshots, from which text is extracted using Optical Character Recognition (Tesseract OCR). The extracted text is preprocessed and vectorized using TF-IDF n-gram representations, then classified into categories including Urgency, Scarcity, Confirmshaming, Social Proof Manipulation, or Not a Dark Pattern, using benchmarked classical machine learning models such as Naive Bayes, Logistic Regression, and LinearSVC. The system is deployed as an interactive Streamlit web application, providing consumers, researchers, and regulators with an accessible tool for auditing manipulative design practices and promoting transparency in digital interfaces.

2. INTRODUCTION & BACKGROUND

As digital commerce has grown, so has the sophistication of interface design techniques aimed at steering user behaviour, often at the expense of informed consent. Researchers and regulators have termed these manipulative design strategies “dark patterns” — user interface choices that benefit an online service by coercing, steering, or deceiving users into making decisions they would not otherwise make. Common examples include countdown timers that reset or are entirely fabricated, claims of limited stock that do not reflect real inventory, reject buttons phrased to induce guilt (“confirmshaming”), and fake notifications suggesting high demand.

Regulatory bodies, including the FTC in the United States and consumer protection authorities in the European Union and India, have begun actively investigating and penalizing such practices. However, manual auditing of websites for dark patterns is slow, labour-intensive, and does not scale to the size of the modern web. This creates a clear need for automated, machine-learning-driven detection systems that can flag manipulative text at scale, either for regulatory auditing, academic research, or direct consumer protection through browser-level tools.

This project addresses that need by combining Natural Language Processing (NLP) with Optical Character Recognition (OCR), allowing the system to analyze not only raw text but also manipulative copy embedded directly within UI screenshots and promotional banners — a format in which dark patterns frequently appear but which most existing text-only detection tools cannot process.

3. SYSTEM ARCHITECTURE & WORKFLOW

The system follows a modular pipeline architecture consisting of input acquisition, text extraction, preprocessing, feature engineering, classification, and result presentation. The workflow is illustrated step-by-step below.

Step 1: Input Acquisition — The user provides input through one of two modes on the Streamlit interface: (a) directly pasted text (review copy, banner text, disclaimers), or (b) an uploaded screenshot of a web/e-commerce interface.

Step 2: OCR Text Extraction (Visual Mode) — If a screenshot is uploaded, the image is preprocessed (grayscale conversion, thresholding, noise removal using Pillow) and passed through Tesseract OCR (via pytesseract) to extract raw text embedded in the interface.

Step 3: Text Preprocessing — Extracted or pasted text undergoes lowercasing, tokenization, punctuation and special-character cleaning, and stop-word removal using NLTK and scikit-learn utilities.

Step 4: Feature Extraction — Cleaned text is converted into numerical feature vectors using TF-IDF vectorization with unigram-to-trigram (1–3 gram) ranges, capturing both individual manipulative keywords and short manipulative phrases.

Step 5: Multi-Class Classification — The feature vectors are passed to a trained classification model. Multiple algorithms — Multinomial Naive Bayes, Logistic Regression (L2-regularized), LinearSVC, and Random Forest — are benchmarked, and the best-performing model is selected for deployment based on macro F1-score.

Step 6: Prediction & Explainability — The selected model outputs the predicted dark-pattern category (Urgency, Scarcity, Confirmshaming, Social Proof Manipulation, or Not a Dark Pattern) along with a confidence score, and highlights the specific trigger keywords/n-grams that most influenced the classification.

Step 7: Report & Visualization — Results are displayed on the Streamlit dashboard with colour-coded severity indicators, a confidence bar, and highlighted trigger phrases, allowing the user to immediately understand why a given text was flagged.

The model training pipeline (data loading, preprocessing, vectorization, training, and evaluation) is developed and validated offline using the Princeton-EC Dark Patterns Dataset and supplementary Kaggle deceptive-patterns data, then serialized (via Joblib) for fast inference within the deployed Streamlit application.

4. KEY DELIVERABLES

- A cleaned and labeled dataset combining the Princeton-EC Dark Patterns Dataset and Kaggle Deceptive Patterns Dataset.
- A trained and benchmarked multi-class text classification model (Naive Bayes, Logistic Regression, LinearSVC, Random Forest) with comparative evaluation.
- An OCR-based text extraction module for screenshot/image inputs using PyTesseract and Pillow.
- A functional Streamlit web application supporting dual-mode input (text and screenshot), real-time classification, confidence scoring, and trigger-keyword highlighting.
- Evaluation report covering Precision, Recall, Macro F1-score, and Confusion Matrix analysis across all target categories.
- Public deployment of the application on Streamlit Community Cloud / Hugging Face Spaces.
- Complete project documentation, including this proposal, system design, and final project report.

5. FUTURE ENHANCEMENTS

- Browser extension integration for real-time, in-page dark pattern flagging while users browse live websites.
- Fine-tuning a lightweight transformer model (e.g., DistilBERT) to improve classification accuracy over purely classical ML approaches.
- Expansion to additional dark pattern categories (e.g., forced continuity, hidden costs, trick questions, bait-and-switch).
- Multilingual detection support for non-English e-commerce interfaces.
- Automated web crawling module to batch-audit entire e-commerce sites rather than single screenshots.
- Integration with regulatory/compliance dashboards for institutional or research-level auditing at scale.
- Severity scoring model that weighs detected patterns by potential consumer harm, not just category.